

An abacus bound on the Heisenberg-crystal depth:

$$|\kappa(\lambda)| \leq k' \text{ for } \lambda \vdash ek' \text{ with empty } e\text{-core}$$

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Abstract

Let $e \geq 2$ and $k' \geq 1$, and let $\kappa: \Pi \rightarrow \Pi$ be the Heisenberg-crystal image map of Gerber ([1], Proposition 3.12) on the level-1 Fock space of $\widehat{\mathfrak{sl}}_e$, computed at charge $s = 0$. We prove that for every partition λ of $n = ek'$ with empty e -core — equivalently, every λ in the support of $v_{k',e} := P_e^{k'}|\emptyset\rangle$ — the Heisenberg-crystal depth $|\kappa(\lambda)|$ satisfies

$$|\kappa(\lambda)| \leq k'.$$

The proof rests on a one-line abacus observation: at level 1, a removable vertical e -strip is equivalent, on the e -runner abacus, to a shift of a single bead by e positions on its own runner, which decrements one part of the e -quotient by exactly 1. Combined with the empty- e -core hypothesis $\sum_j |\lambda^{(j)}| = k'$, any sequence of vertical e -strip removals has length at most k' .

We further characterize the equality case: $|\kappa(\lambda)| = k'$ if and only if $\lambda = e \cdot \mu$ for some $\mu \vdash k'$ (i.e., each part of λ has multiplicity divisible by e). The forward direction is proved directly from reversibility; the converse follows by combining this with Gerber's Young-graph identification of the empty-source component ([1], Proposition 3.12(2)).

Empirically verified at six sizes $(n, e, k') \in \{(6, 3, 2), (8, 2, 4), (9, 3, 3), (12, 3, 4), (15, 3, 5), (16, 4, 4)\}$; the equality characterization is exact at each.

1 Introduction and statement

1.1 Background

Gerber's paper [1] on *Heisenberg algebra, wedges and crystals* introduces a commuting family of *Heisenberg* crystal operators $\{\tilde{b}_{a,c}^{\pm 1}\}_{a \in \mathbb{Z}_{>0}, c \in \mathbb{Z}}$ on the level- ℓ higher-level q -Fock space. At level $\ell = 1$ and charge $s = 0$ (the setting of this note), this action lifts to a crystal on the usual level-1 Fock space $\mathcal{F}$ of $\widehat{\mathfrak{sl}}_e$, whose basis is indexed by partitions.

Two structural facts about this crystal, proved in [1, Prop. 3.12], are crucial:

- (i) Each connected component of the H-crystal is graded and isomorphic (as a graded graph) to the Young graph $\mathcal{Y}$.
- (ii) The depth of a basis vector $|\lambda\rangle$ inside its component equals $|\kappa(\lambda)|$, where $\kappa(\lambda)$ is a distinguished partition (the image of λ under the map to the Young graph).

The image partition $\kappa(\lambda)$ is not readily accessible in closed form: the constructive recipe ([1, Thm. 5.11], level-1 specialization) iteratively removes *good vertical e -strips* from λ , and $|\kappa(\lambda)|$ is the number of such removals until an H-highest-weight vertex is reached.

1.2 The theorem

Write Π_k for the set of partitions of k , and $\lambda^{(0)}, \dots, \lambda^{(e-1)}$ for the parts of the e -quotient of λ .

Theorem 1 (H-crystal depth bound). *Let $e \geq 2$ and $k' \geq 1$. For every partition λ of $n = ek'$ with empty e -core,*

$$|\kappa(\lambda)| \leq k'.$$

Moreover, equality $|\kappa(\lambda)| = k'$ holds if and only if $\lambda = e \cdot \mu$ for some $\mu \vdash k'$, where $e \cdot \mu$ denotes the partition in which each part of μ appears e times.

1.3 Context (post-hoc)

This bound is the residue of a stronger attempted claim $|\kappa(v_{k',e})| = k'$ (with $v_{k',e} = \sum_{\lambda} \chi_{(ek')}^{\lambda} |\lambda\rangle$), which fails category-theoretically: Gerber's crystal is defined on basis vectors, whereas $v_{k',e}$ is a linear combination spread across many H-components (see the note [for-robin/2026-08-11-wake-q21-q23-downgraded](#) for detail). What survives, and is proved here as a structural constraint on the support of $v_{k',e}$, is the uniform depth bound.

2 Preliminaries

2.1 The abacus and the e -quotient

Fix a partition $\lambda = (\lambda_1 \geq \lambda_2 \geq \dots)$ and an integer $N \geq \ell(\lambda)$ (where $\ell(\lambda)$ denotes the number of nonzero parts). The *beta-numbers of λ with N beads* are

$$\beta_i := \lambda_i + (N - i), \quad i = 1, 2, \dots, N.$$

These are N distinct non-negative integers, strictly decreasing: $\beta_1 > \beta_2 > \dots > \beta_N \geq 0$.

Place *beads* on the semi-infinite number line $\mathbb{Z}_{\geq 0}$ at positions $\beta_1, \dots, \beta_N$; call this the *e -abacus of λ with N beads*. Position $p \in \mathbb{Z}_{\geq 0}$ lies on *runner* $p \bmod e$, at *height* $\lfloor p/e \rfloor$. Write $N_j := |\{i : \beta_i \equiv j \pmod{e}\}|$ for the number of beads on runner j .

Remark 2. *Different choices of N give abaci with the same λ but differing bead counts on each runner; the differences are bookkeeping and do not affect any argument below.*

For each runner $j \in \{0, 1, \dots, e-1\}$, let the beads on runner j lie at heights $h_{j,1} > h_{j,2} > \dots > h_{j,N_j} \geq 0$. The j -th part of the *e -quotient of λ* is the partition $\lambda^{(j)}$ with parts

$$\lambda_i^{(j)} := h_{j,i} - (N_j - i), \quad i = 1, \dots, N_j.$$

(These are non-negative and weakly decreasing.) The *e -core of λ* is the partition obtained by pushing every bead maximally left on its own runner; equivalently, all $\lambda^{(j)} = \emptyset$ iff λ equals its own e -core iff every runner's beads occupy positions $\{j, j+e, j+2e, \dots, j+(N_j-1)e\}$.

We record two standard facts (proved e.g. in [2], Chapter 2):

Lemma 3 (Standard). $|\lambda^{(j)}| = \sum_{i=1}^{N_j} h_{j,i} - \binom{N_j}{2}$.

Lemma 4 (Standard). $|\lambda| = |\text{core}_e(\lambda)| + e \cdot \sum_{j=0}^{e-1} |\lambda^{(j)}|$.

2.2 Level-1 vertical e -strips

Rows and columns are 0-indexed: cell (r, c) sits in row r , column c . The *content* of cell (r, c) is $c - r$, its residue $(c - r) \bmod e$.

Gerber's Definition 5.1 defines a *vertical e -strip* (in the level- ℓ setting) as a sequence of e boxes $\gamma_1, \dots, \gamma_e$ with no consecutive pair in the same row ("no horizontal domino"), and $c(\gamma_{i+1}) = c(\gamma_i) - 1$ for all i . At level 1 the ordering tag on boxes is trivial; the constraint "distinct rows plus decreasing contents by 1" forces $\gamma_i = (r + i - 1, c)$ for a single common column c and starting row r . Thus:

Proposition 5 (Level-1 vertical e -strip). *At level 1, a vertical e -strip is a set of e cells in a single column at e consecutive rows, i.e., $\{(r, c), (r + 1, c), \dots, (r + e - 1, c)\}$ for some $r \geq 0, c \geq 0$.*

A vertical e -strip is *removable from* λ if every cell $(r + i, c)$ ($0 \leq i < e$) is the rightmost cell of its row in λ (so $\lambda_{r+i+1} = c + 1$ using 1-indexed parts, i.e. $\lambda_{r+1} = \dots = \lambda_{r+e} = c + 1$), and the removal (i.e., decreasing each of these e parts by 1) yields a partition (equivalently, if row $r + e + 1$ exists then $\lambda_{r+e+1} \leq c$).

2.3 Gerber's Heisenberg crystal at level 1

Following [1, §3.2 and Thm. 5.11] in the level-1 specialization, the H-crystal descent operator $\tilde{b}_{1,c}^{-1}$ on the basis $\{|\lambda\rangle\}$ acts by removing the k -th *good removable vertical e -strip* of λ (for an appropriate index k selected by the content c). The definition of *good removable* follows the greedy rule of [1, Def. 5.5]: order the removable vertical e -strips by the strict content-lex order $\succ$ (with top cell of each strip having largest content); the good strips $X_1 \succ X_2 \succ \dots$ are picked greedily as X_j is the $\succ$ -maximum removable strip disjoint from $X_1, \dots, X_{j-1}$.

A basis vector $|\lambda\rangle$ is *H-highest-weight* iff no vertical e -strip is removable from λ (equivalently, λ has no e consecutive equal rows). By [1, Prop. 3.12(4)], the H-depth $|\kappa(\lambda)|$ equals the number of applications of the descent operator (any legal sequence of good removals) required to reach an H-highest-weight vertex.

Remark 6. *The upper bound argument below does not require any property of the "good" selection: it uses only that each step is some vertical e -strip removal, and hence decrements the quotient-size sum by exactly one.*

3 Key lemma: strip removal = single bead shift

Lemma 7 (Abacus correspondence for vertical e -strip removal). *Let λ be a partition with a removable vertical e -strip S at column c , rows $r, r + 1, \dots, r + e - 1$. Let $\lambda' := \lambda \setminus S$. Fix $N \geq \ell(\lambda)$; use N beads for both abaci. Then the bead multiset of λ' is obtained from that of λ by removing one bead at position $c + N - r$ and adding one bead at position $c + N - r - e$. In particular, both positions lie on runner $(c - r) \bmod e$, and the change is a single bead shift by e positions leftward on that runner.*

Proof. Using 1-indexed row indices, removability of the strip gives $\lambda_{r+1} = \lambda_{r+2} = \dots = \lambda_{r+e} = c + 1$, and either $r + e = \ell(\lambda)$ or $\lambda_{r+e+1} \leq c$.

The beta-numbers of λ satisfy, for $i = 1, \dots, e$,

$$\beta_{r+i} = \lambda_{r+i} + (N - r - i) = (c + 1) + (N - r - i) = c + N - r - i + 1.$$

So $\beta_{r+1} = c + N - r$, $\beta_{r+2} = c + N - r - 1$, $\dots$, $\beta_{r+e} = c + N - r - e + 1$; the block $\{\beta_{r+1}, \dots, \beta_{r+e}\}$ is exactly the set of e consecutive integers $\{c + N - r - e + 1, \dots, c + N - r\}$.

Removing S decreases each λ_{r+i} ($i = 1, \dots, e$) by 1, hence each β_{r+i} by 1. The new bead positions $\{\beta'_{r+1}, \dots, \beta'_{r+e}\}$ are the shifted block $\{c + N - r - e, \dots, c + N - r - 1\}$.

The symmetric difference (old block) $\triangle$ (new block) = one removal of $c + N - r$ and one addition of $c + N - r - e$; the other $e - 1$ integers are common. All other β_j ($j \notin \{r + 1, \dots, r + e\}$) are unchanged.

Both positions on runner $(c - r) \bmod e$: both differ by a multiple of e , and $c + N - r \equiv c - r \pmod{e}$ (since N and $N - e$ differ by e ; more directly $(c + N - r) - (c + N - r - e) = e$).

Position $c + N - r$ is a bead of λ : it equals β_{r+1} by construction.

Position $c + N - r - e$ is not a bead of λ : we must show $c + N - r - e \notin \{\beta_1, \dots, \beta_N\}$. It equals $\beta_{r+e} - 1$. In the block $\{\beta_{r+1}, \dots, \beta_{r+e}\}$ (values $c + N - r - e + 1, \dots, c + N - r$), it is one below the minimum, so absent. Above the block, $\beta_{r+e-1} = c + N - r - e + 2 > c + N - r - e$. Below the block, either $r + e = \ell(\lambda) = N$ (no further beads with index $> r + e$ having positive λ -part; but $\beta_j = N - j$ for $j > \ell(\lambda)$, so $\beta_{r+e+1} = N - r - e - 1 = c + N - r - e - (c + 1) \leq c + N - r - e - 1$) or $\lambda_{r+e+1} \leq c$, giving $\beta_{r+e+1} = \lambda_{r+e+1} + (N - r - e - 1) \leq c + N - r - e - 1$. In both cases $\beta_{r+e+1} \leq c + N - r - e - 1 < c + N - r - e$, and subsequent β_j ($j > r + e + 1$) are strictly smaller. So $c + N - r - e \notin \{\beta_1, \dots, \beta_N\}$, as claimed.

Combining: the bead multiset of λ' is $(\{\beta_1, \dots, \beta_N\} \setminus \{c + N - r\}) \cup \{c + N - r - e\}$. This is a single bead moving from position $c + N - r$ to $c + N - r - e$ on the runner $(c - r) \bmod e$, leftward by e positions (i.e., height decreases by 1). $\square$

Corollary 8 (Strip removal decrements quotient-size sum by exactly 1). *Under the hypotheses of Lemma 7, with $j := (c - r) \bmod e$,*

$$|\lambda'^{(j)}| = |\lambda^{(j)}| - 1, \quad |\lambda'^{(j')}| = |\lambda^{(j')}| \text{ for } j' \neq j.$$

Consequently $\sum_{i=0}^{e-1} |\lambda'^{(i)}| = \sum_{i=0}^{e-1} |\lambda^{(i)}| - 1$, and $\text{core}_e(\lambda') = \text{core}_e(\lambda)$.

Proof. By Lemma 7, exactly one bead on runner j shifts from height h to height $h - 1$; other runners are untouched. By Lemma 3, $|\lambda^{(j)}|$ is (sum of heights on runner j) minus $\binom{N_j}{2}$; the bead-count N_j is unchanged; the sum of heights decreases by 1. Hence $|\lambda'^{(j)}| = |\lambda^{(j)}| - 1$.

The e -core depends only on the bead-count profile $(N_0, \dots, N_{e-1})$, which is preserved by any single-runner shift. So the core is preserved. $\square$

4 Proof of Theorem 1, upper bound

Assume $\lambda \vdash n = ek'$ has empty e -core. By Lemma 4,

$$ek' = |\lambda| = 0 + e \cdot \sum_{j=0}^{e-1} |\lambda^{(j)}|,$$

so $\sum_j |\lambda^{(j)}| = k'$.

By [1, Prop. 3.12(4)] (level-1 specialization), $|\kappa(\lambda)|$ is the length of the good-removal chain from λ to an H-highest-weight vertex. Denote this chain

$$\lambda = \lambda^{[0]} \rightarrow \lambda^{[1]} \rightarrow \dots \rightarrow \lambda^{[m]},$$

where $\lambda^{[k]} \rightarrow \lambda^{[k+1]}$ is a removal of a vertical e -strip (the $\lambda^{[k+1]}$ is a specific “good” strip removal per Def. 5.5, but this is not needed for the bound), $m = |\kappa(\lambda)|$, and $\lambda^{[m]}$ is H-highest-weight (no removable strips).

By Corollary 8, each step decrements $\sum_j |\lambda^{(j)}|$ by exactly 1 and preserves the (empty) e -core. So for each $k \leq m$,

$$\sum_{j=0}^{e-1} |\lambda^{[k],(j)}| = k' - k.$$

Since the left side is a sum of non-negative integers, $k' - k \geq 0$, giving $k \leq k'$ for all k along the chain, and in particular $m \leq k'$. This proves $|\kappa(\lambda)| \leq k'$. $\square$

5 Equality case

We now identify the partitions saturating the bound. The forward direction (“if $|\kappa| = k'$ then $\lambda = e \cdot \mu$ ”) is proved directly from Lemma 7; the converse relies on [1, Prop. 3.12(2)].

5.1 Forward direction

Proposition 9. *If $\lambda \vdash ek'$ has empty e -core and $|\kappa(\lambda)| = k'$, then $\lambda = e \cdot \mu$ for some $\mu \vdash k'$ (i.e., every part of λ appears with multiplicity divisible by e).*

Proof. Suppose $|\kappa(\lambda)| = k'$. By the analysis of Section 4, the good-removal chain $\lambda = \lambda^{[0]} \rightarrow \cdots \rightarrow \lambda^{[k']}$ has length exactly k' , and $\sum_j |\lambda^{[k'],(j)}| = 0$. So $\lambda^{[k']}$ has empty e -quotient AND empty e -core, hence equals $\emptyset$ (by Lemma 4).

Reversing the chain: λ is obtained from $\emptyset$ by successively *adding* k' vertical e -strips. At level 1 (Proposition 5) each added strip consists of e cells in a single column. Hence for every column $c \geq 0$, the number of cells of λ in column c is $e \cdot (\text{number of strips added at column } c)$; in particular, every column height of λ is divisible by e .

Column heights of λ are the multiplicities of parts of λ' (the conjugate). Concretely, if the columns of λ have heights $e \cdot m_1 \geq e \cdot m_2 \geq \cdots \geq e \cdot m_\ell$ (with $m_i \geq 1$), then $\lambda' = (em_1, em_2, \dots, em_\ell)$, and dually λ has each part appearing with multiplicity a multiple of e . Explicitly: if column c has height et_c ($t_c \geq 0$), then the parts of λ equal to j (for each $j \geq 1$) number $e(t_{j-1} - t_j) \geq 0$, a non-negative multiple of e .

Set μ to be the partition with parts j appearing with multiplicity $t_{j-1} - t_j$; then $\lambda = e \cdot \mu$, and $|\mu| = \frac{1}{e}|\lambda| = k'$. $\square$

5.2 Converse direction

Proposition 10. *For every $\mu \vdash k'$, $|\kappa(e \cdot \mu)| = k'$.*

Proof. By [1, Prop. 3.12(1),(2)], the connected component of $|\emptyset\rangle$ in the H-crystal is a graded, connected graph isomorphic to the Young graph $\mathcal{Y}$; in particular, the number of vertices at depth exactly k' inside this component equals $|\Pi_{k'}|$, the number of partitions of k' .

Every such depth- k' vertex $|\lambda\rangle$ satisfies $|\kappa(\lambda)| = k'$ (by [1, Prop. 3.12(4)]). By Proposition 9, each such λ is of the form $e \cdot \mu'$ for some $\mu' \vdash k'$. Since the map $\mu' \mapsto e \cdot \mu'$ is an injection from $\Pi_{k'}$ to partitions, the depth- k' vertices form a subset of the $|\Pi_{k'}|$ -element set $\{e \cdot \mu' : \mu' \vdash k'\}$.

But this subset has exactly $|\Pi_{k'}|$ elements, so it equals $\{e \cdot \mu' : \mu' \vdash k'\}$. In particular, $|\kappa(e \cdot \mu)| = k'$ for every $\mu \vdash k'$. $\square$

Combining Propositions 9 and 10 completes the proof of Theorem 1.

6 Computational verification

We have implemented the level-1 Heisenberg-crystal descent (following [1, Thm. 5.11]: iterated good vertical e -strip removal) and computed $|\kappa(\lambda)|$ for every partition λ in $\text{supp}(v_{k',e}) = \{\lambda \vdash ek' : \text{core}_e(\lambda) = \emptyset\}$ at six sizes.

(n, e, k')	$ \text{supp}(v_{k',e}) $	$\max \kappa $	$\#\{\lambda : \kappa = k'\}$	Distribution of $ \kappa $
$(6, 3, 2)$	9	2	$2 = \Pi_2 $	$[5, 2, 2]$
$(8, 2, 4)$	20	4	$5 = \Pi_4 $	$[5, 3, 4, 3, 5]$
$(9, 3, 3)$	22	3	$3 = \Pi_3 $	$[10, 5, 4, 3]$
$(12, 3, 4)$	51	4	$5 = \Pi_4 $	$[20, 10, 10, 6, 5]$
$(15, 3, 5)$	108	5	$7 = \Pi_5 $	$[36, 20, 20, 15, 10, 7]$
$(16, 4, 4)$	105	4	$5 = \Pi_4 $	$[51, 22, 18, 9, 5]$

Table 1: Distribution of $|\kappa(\lambda)|$ over $\text{supp}(v_{k',e})$. In every case the upper bound $\max |\kappa| \leq k'$ is tight, and equality is achieved exactly by the $|\Pi_{k'}|$ partitions of the form $e \cdot \mu$. Distribution entry $j = \#\{\lambda \in \text{supp}(v_{k',e}) : |\kappa(\lambda)| = j\}$.

The code lives in `~/projects/probes/2026-08-11-q21-gerber-kappa/(gerber_heisenberg.py, probe_extend.py)`; the raw output for the extended table is in `probe_extend.log`.

7 What is (and is not) proved

- **Proved rigorously here:** Theorem 1 in full, i.e., both the bound $|\kappa(\lambda)| \leq k'$ and the equality characterization $|\kappa(\lambda)| = k' \iff \lambda = e \cdot \mu$. The bound uses only the abacus correspondence (Lemma 7); the equality forward direction uses reversibility; the converse cites Gerber’s Young-graph identification [1, Prop. 3.12(2)].
- **Cited without reproof:** the Young-graph structure of the empty-source H-component (Gerber Prop. 3.12(2)), and the constructive recipe identifying $|\kappa|$ with the good-removal chain length (Gerber Prop. 3.12(4) + Thm. 5.11).
- **Not addressed here:** the value of $\varepsilon_i(v_{k',e})$, Clío’s Conjecture 10 (that $\varepsilon_i(v_{k',e}) = 1$ uniformly) and Conjecture 11 (endpoint decomposition of $\pi(w)$). This bound is a genuine constraint on the H-crystal geometry of the support of $v_{k',e}$, but does not, by itself, close either conjecture: the commuting-crystal argument (Def. 3.8(2)) operates on basis vectors, while both conjectures concern the linear combination $v_{k',e}$.
- **No known gap.** The proof of Theorem 1 is complete modulo the cited external facts.

References

- [1] T. Gerber, *Heisenberg algebra, wedges and crystals*, J. Algebraic Combin. **49**(1) (2019), 99–124. [arXiv:1612.08760](#).
- [2] G. D. James and A. Kerber, *The Representation Theory of the Symmetric Group*, Encyclopedia of Mathematics and its Applications, Vol. 16, Addison-Wesley, 1981.